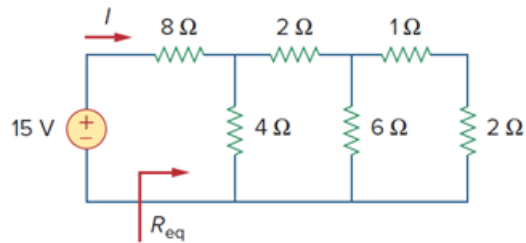


Problem

For the ladder network in Fig. 2.104, find I and R_{eq} .

Figure 2.104:



Step-by-step solution

Step 1 of 7

Consider the following circuit diagram:

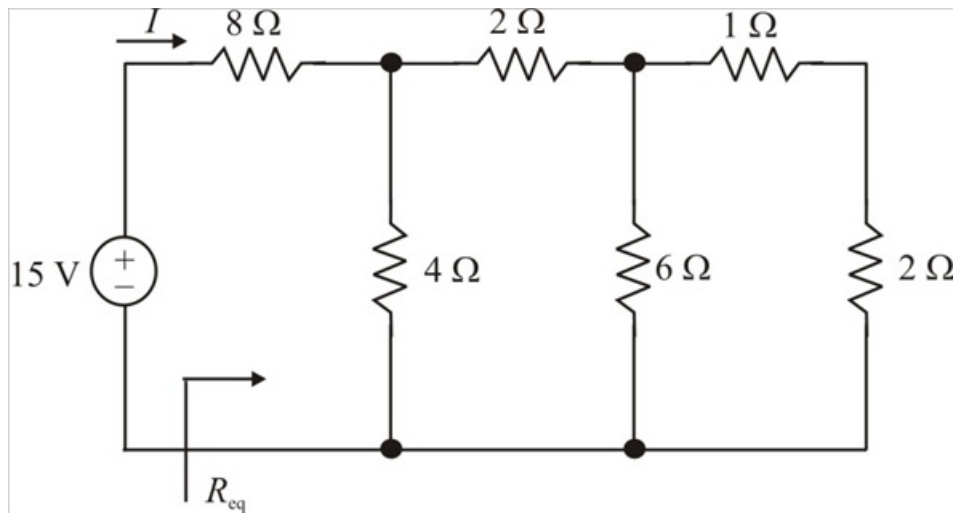


Figure 1

Step 2 of 7

The 1 Ω and 2 Ω resistors are in series. Their combined resistance is,

$$1\Omega + 2\Omega = 3\Omega$$

This 3 Ω and 6 Ω resistors are in parallel. Their combined resistance is,

$$3\Omega \parallel 6\Omega = \frac{3 \times 6}{3 + 6} \\ = 2\Omega$$

Step 3 of 7

Then the reduced circuit is shown in Figure 2.

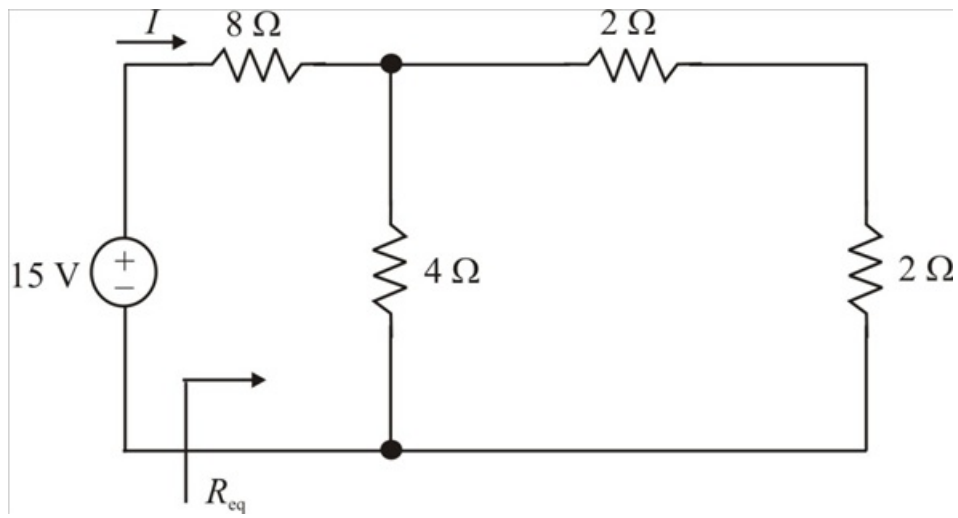


Figure 2

Step 4 of 7

The 2 Ω and 2 Ω resistors are in series. Their combined resistance is,

$$2\Omega + 2\Omega = 4\Omega$$

This 4 Ω and 4 Ω resistors are in parallel. Their combined resistance is,

$$4\Omega \parallel 4\Omega = \frac{4 \times 4}{4 + 4} \\ = 2\Omega$$

Step 5 of 7

Then the reduced circuit is shown in Figure 3.

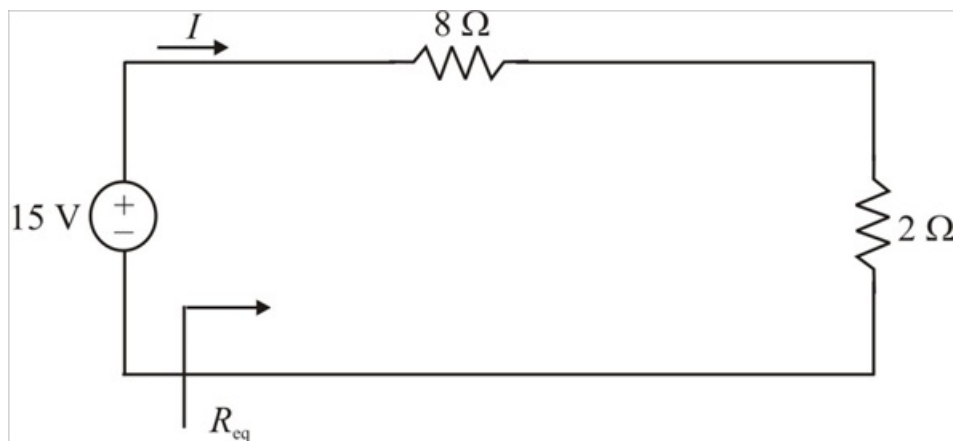


Figure 3

Step 6 of 7

The 8 Ω and 2 Ω resistors are in series. Their combined resistance is:

$$R_{eq} = 8\Omega + 2\Omega \\ = 10\Omega$$

Therefore, the equivalent resistance R_{eq} is 10Ω.

By using Ohms law, the current I can be calculated as follows.

$$\begin{aligned} I &= \frac{V}{R} \\ &= \frac{15}{10} \\ &= 1.5 \text{ A} \end{aligned}$$

Therefore, the value of current I is 1.5 A.